



Capstone Project: Reading Habits

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Table of Contents

01 Goals/Motivation

04 Model Design & Implementation

02 Data Discovery & Discussions

05 Social/Community Impact

03 Exploratory Analysis





01

Goals/Motivation

Goals and Motivations



Goal: Analyze the daily reading habits of about 3,000 individuals and their demographics

Our Interest:

- Thought it would be a more meaningful dataset to work with
- Interesting to see if your demographics can have an effect on your reading habits



Two long bookshelves at the top of the image, filled with books of various colors (red, green, blue, orange, purple).A small lantern with a lit candle inside, positioned on the left side of the central area.A small, open purple book floating in the air on the left side.A small, open red book floating in the air on the right side.A stack of four books (blue, orange, blue, orange) on the bottom left.A stack of four books (blue, orange, blue, orange) on the bottom right.

02

Data Discovery & Discussions

Dataset

From Kaggle: (Source - PEW Research Center)

<https://www.kaggle.com/datasets/vipulgote4/reading-habit-dataset?resource=download>

Reading Habits

Data Items:

- Demographics (ie. age, race, sex, income, etc.)
- Number of Books Read
- Type of Writing the Participants Read
 - Ex: Read any audiobooks in the last 12 months?
- 2832 entries
- Across 12 months



Issues

Credibility?

- Says it is from the Pew Research Center but doesn't provide a link
- Don't know how they chose group of people (majority participants seem to be White)
- May have lack of diversity in data

White	78%
Black or African-...	12%
Other (268)	9%

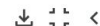
Hard to Manipulate

- Very little of the dataset is in numerical value
- Had to convert Strings to numerical values for the model diagram
- Lead to a lack of variety in exploratory analysis graphs (not the type of data for scatter plots/line charts)

Incomes
\$20,000 to under \$30,000
Less than \$10,000
Less than \$10,000

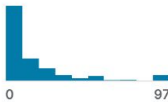
Screenshot

BigML_Dataset_5f50a62795a9306aa200003e.csv (411.39 kB)



Detail Compact Column

10 of 14 columns ▾

# Age age	△ Sex sex	△ Race race	△ Marital status? marital status	△ Education education	△ Employment full time/ retired/ part time	△ Incomes categorical values	# How many books ... How many books did you read during last 12months?	△ F Read dur
 16 93	Female 52% Male 48%	White 78% Black or African-... 12% Other (268) 9%	Married 54% Never been marri... 21% Other (708) 25%	High school grad... 24% Some college, no... 23% Other (1493) 53%	Employed full-time 44% Retired 21% Other (989) 35%	\$100,000 to unde... 19% \$50,000 to under... 14% Other (1908) 67%	 0 97	Yes [nu Oth
66	Male	Refused	Divorced	College graduate	Retired	\$20,000 to under \$30,000	97	Yes
46	Male	Native American/American Indian	Married	High school graduate	Employed full-time	Less than \$10,000	97	Yes
32	Male	Mixed race	Never been married	High school graduate	Employed full-time	Less than \$10,000	97	No
27	Male	Mixed race	Married	High school graduate	Employed full-time	\$40,000 to under \$50,000	97	Yes
16	Female	Mixed race	Never been married	High school incomplete	Employed part-time	\$10,000 to under \$20,000	97	Yes
55	Female	Asian or Pacific Islander	Divorced	Some college, no 4- year degree	Have own business/self- employed	\$40,000 to under \$50,000	97	Yes
63	Male	Black or African- American	Widowed	Some college, no 4- year degree	Employed full-time	\$50,000 to under \$75,000	97	Yes

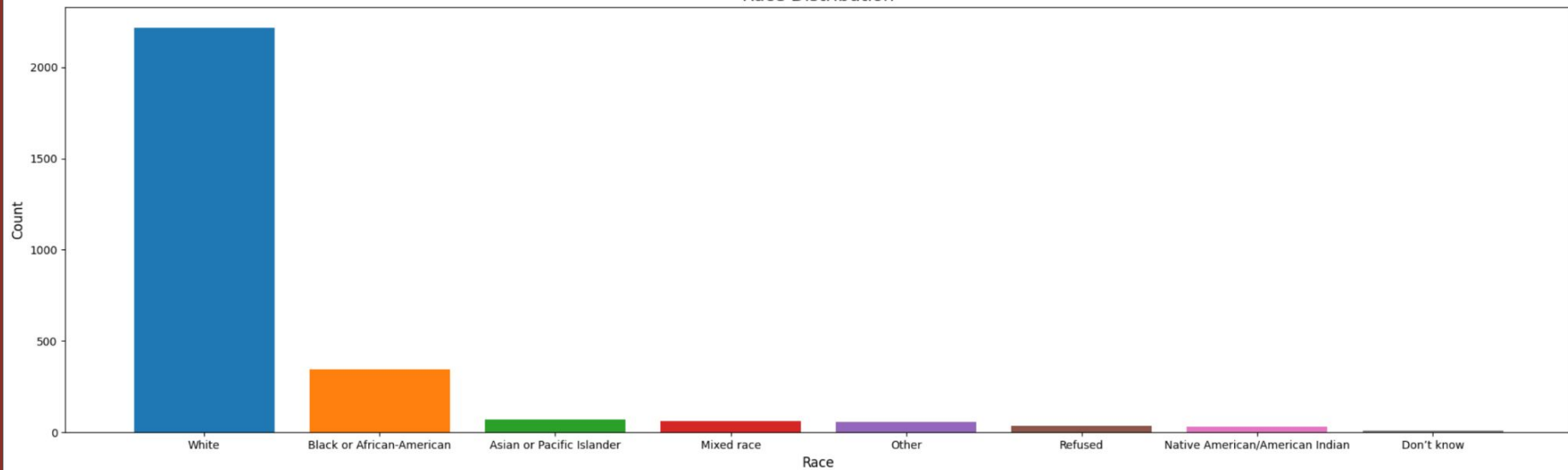


03

Exploratory Analysis

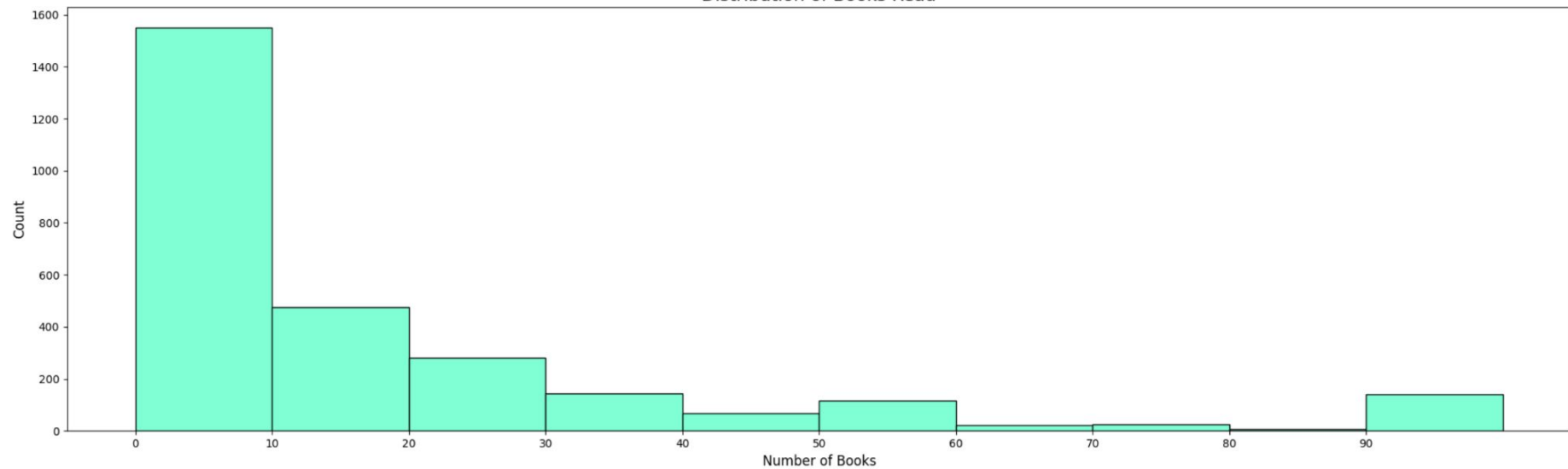


Race Distribution

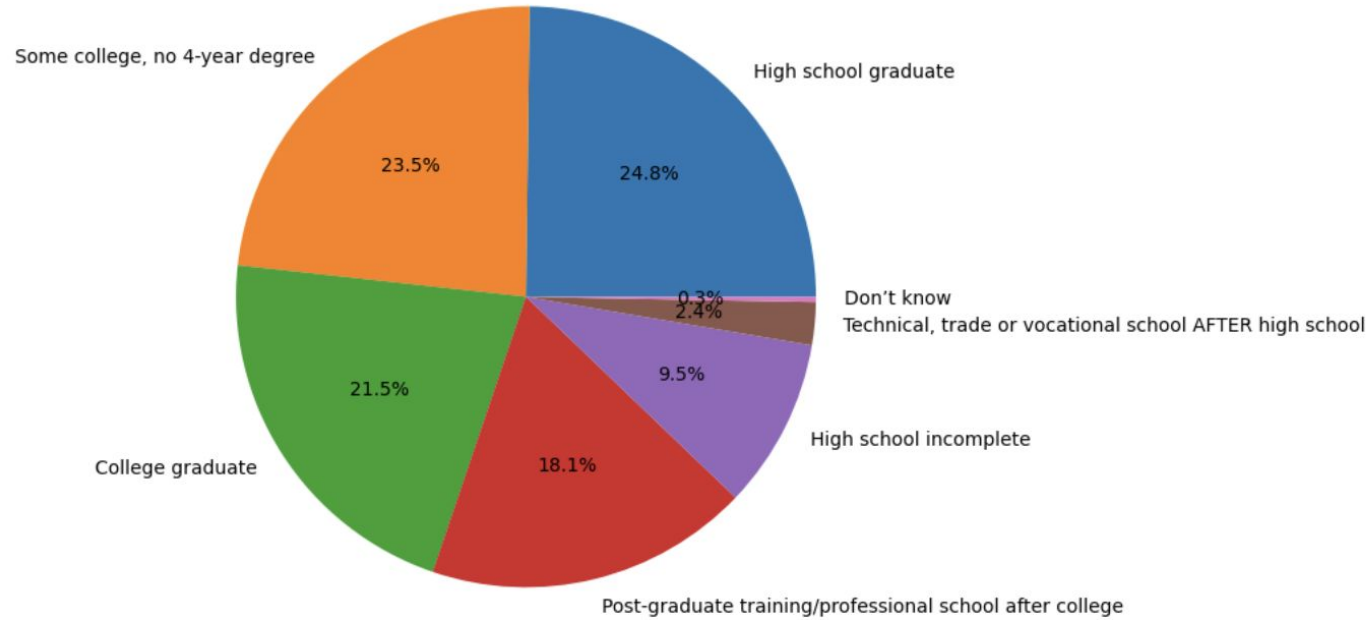




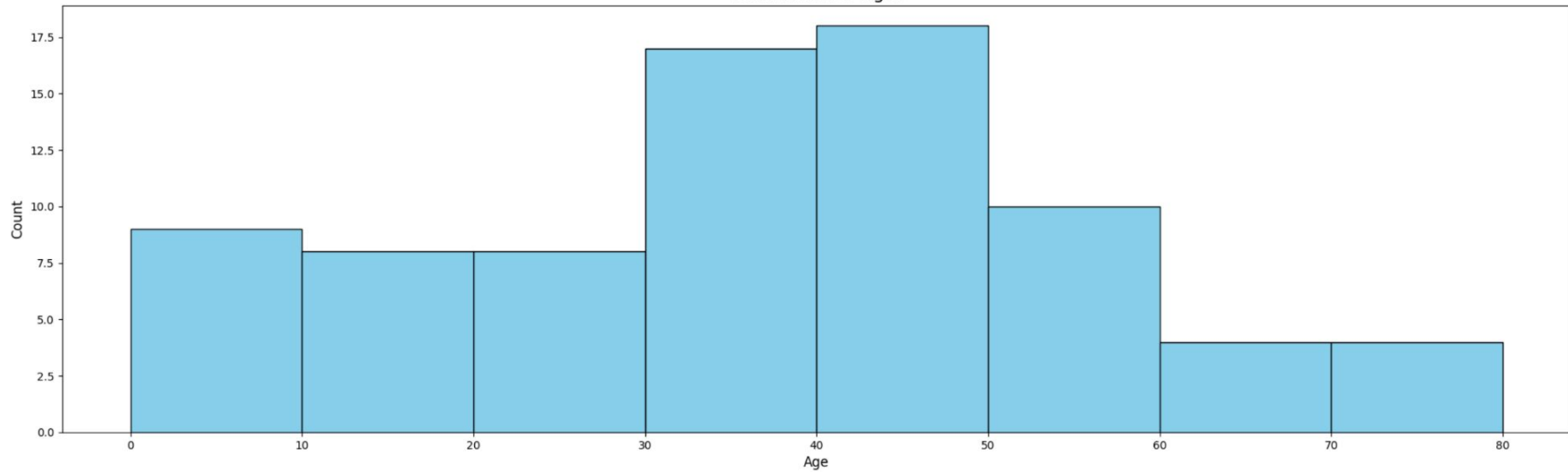
Distribution of Books Read



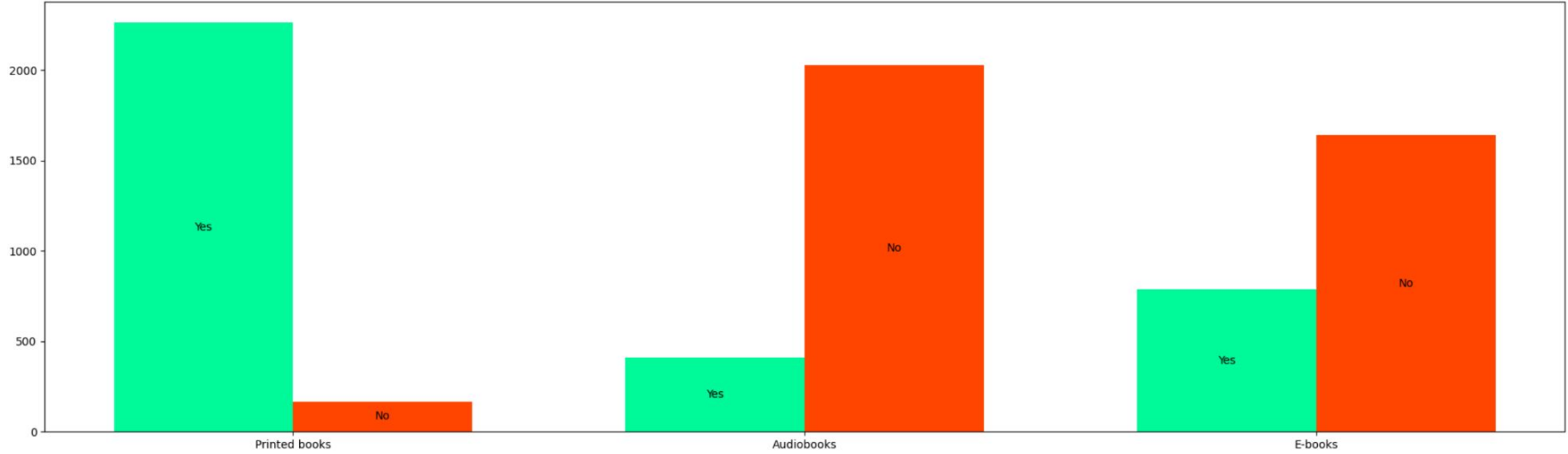
Education Distribution



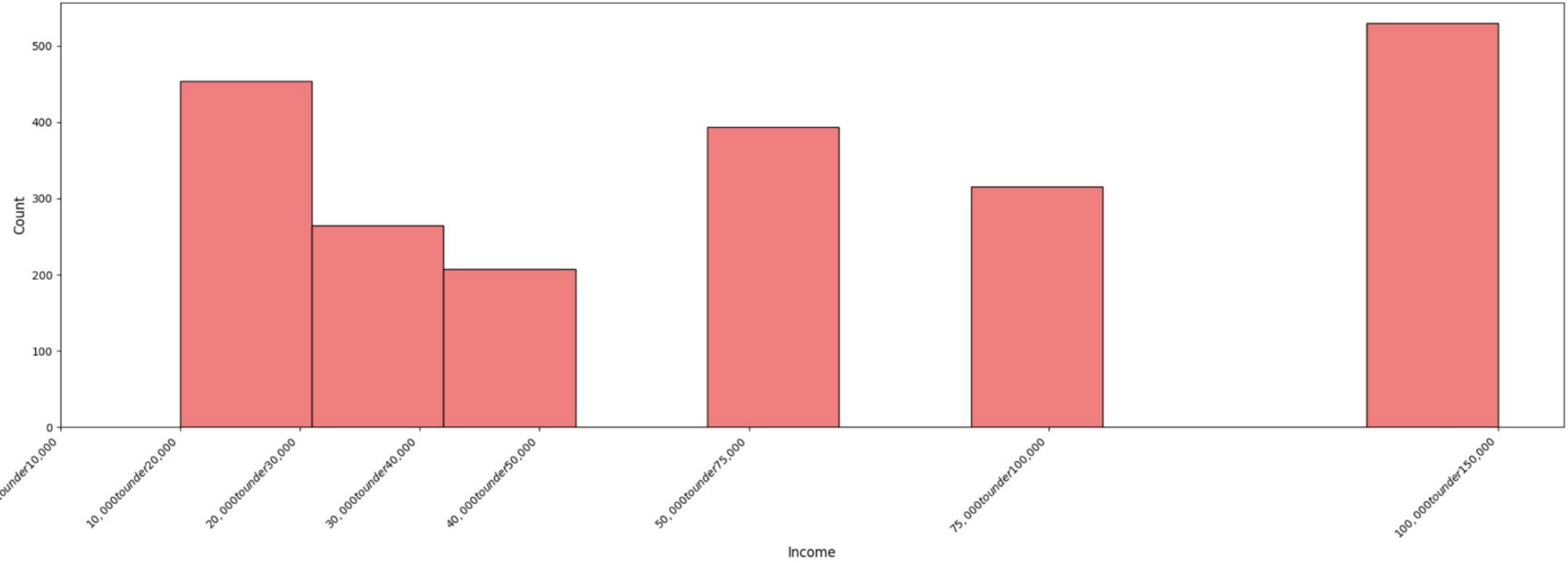
Distribution of Ages

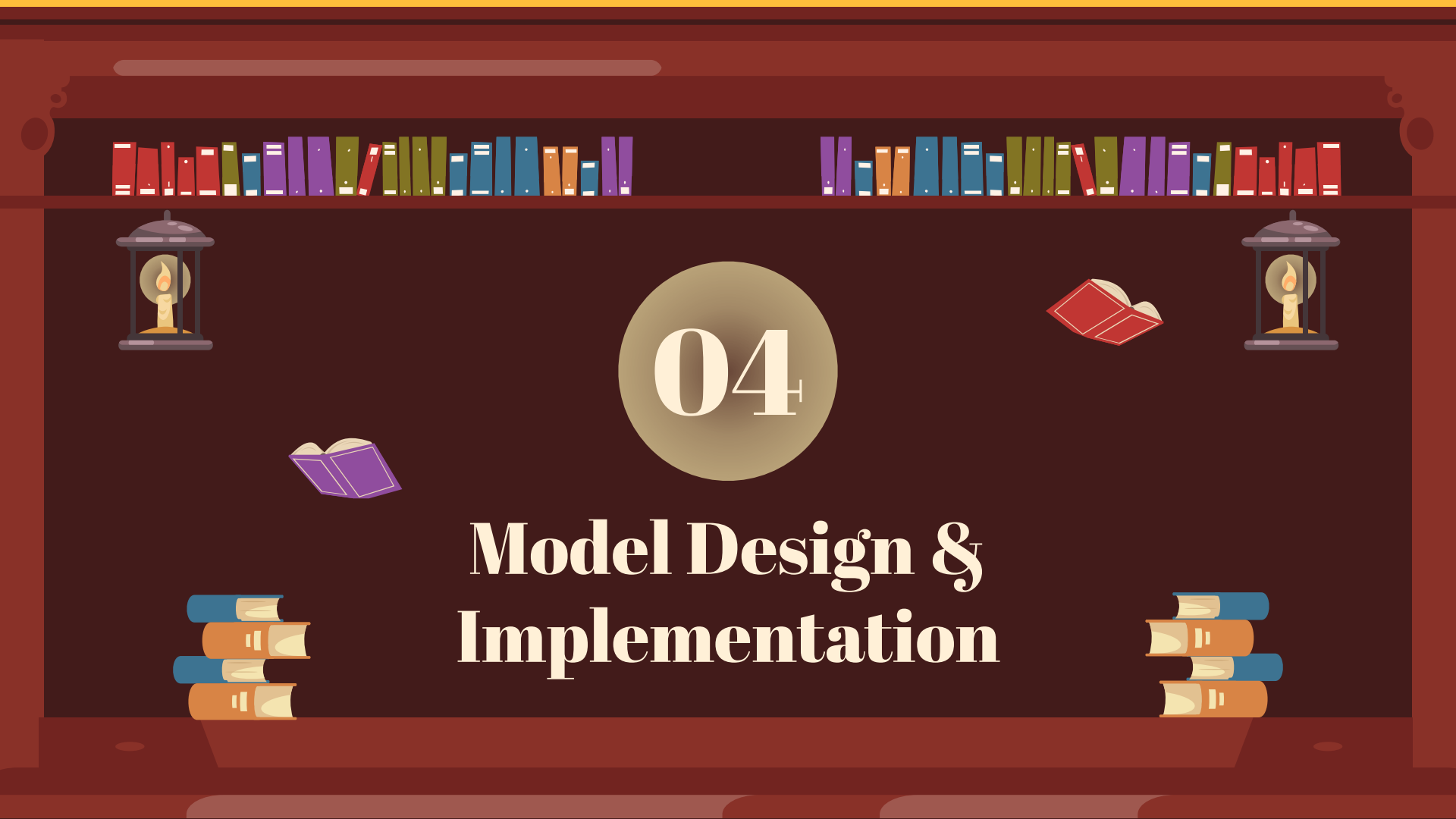


Have You Read This Type of Writing?



Distribution of Incomes





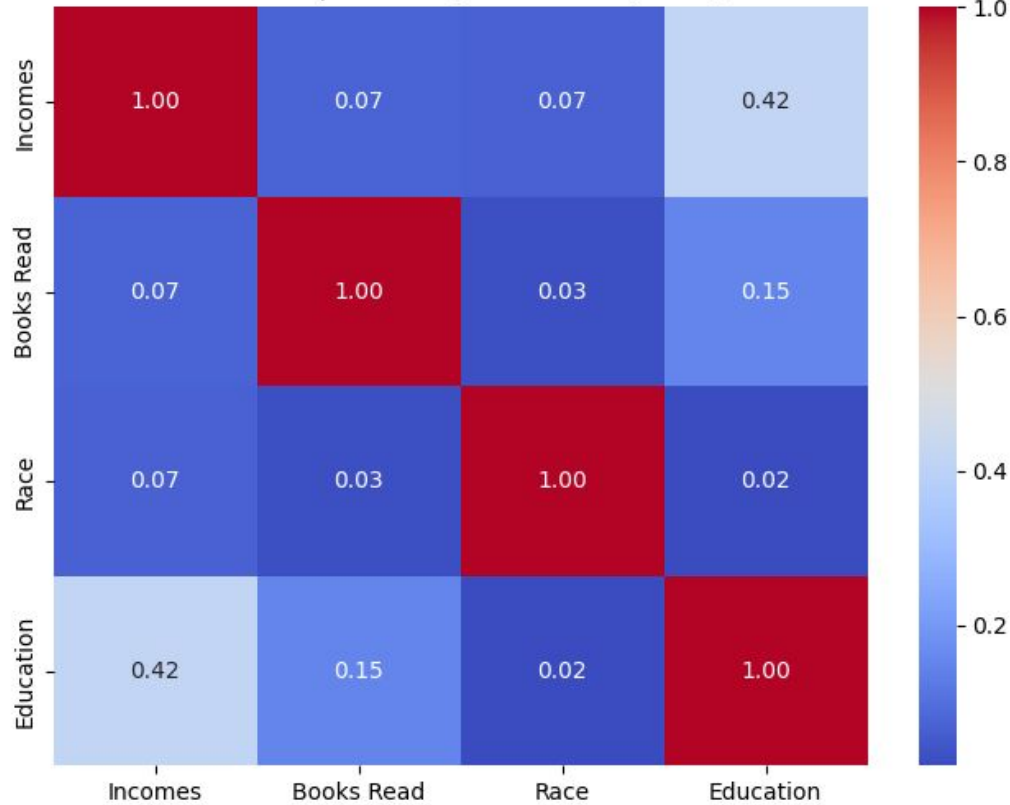
04

Model Design & Implementation

Correlation Model

- Heat map (Statistical Model)
- Visualize the correlation between # of books read and other possible affecting variables
 - Race
 - Education
 - Income
- Have to give education and race a numerical value
 - Education based on hierarchy of degree
 - Race value is random

Correlation Heatmap: Income, Books Read, Race, Education



```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Read dataset
df = pd.read_csv('reading_habits.csv')

# Map income ranges to midpoints
income_mapping = {
    '$0 to under $10,000': 5000,
    '$10,000 to under $20,000': 15000,
    '$20,000 to under $30,000': 25000,
    '$30,000 to under $40,000': 35000,
    '$40,000 to under $50,000': 45000,
    '$50,000 to under $75,000': 62500,
    '$75,000 to under $100,000': 87500,
    '$100,000 to under $150,000': 125000,
    '$150,000 to under $200,000': 175000,
    '$200,000 or more': 200000
}

df['Incomes'] = df['Incomes'].map(income_mapping)

# Rank education levels
education_mapping = {
    'Don't know': 0,
    'High school incomplete': 1,
    'High school graduate': 2,
    'Some college, no 4-year degree': 3,
    'Technical, trade or vocational school AFTER high school': 4,
    'College graduate': 5,
    'Post-graduate training/professional school after college': 6
}
```

```
df['Education'] = df['Education'].map(education_mapping)

# Encode 'Race' (factorize turns categories into integers)
df['Race'] = pd.factorize(df['Race'])[0]

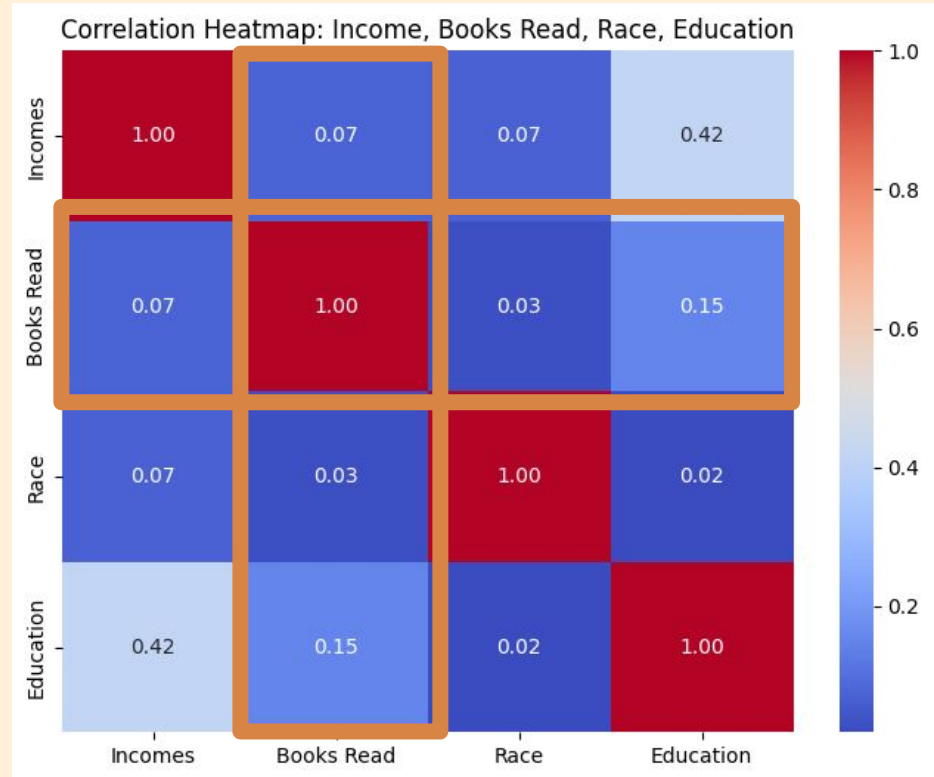
# Select columns and rename 'Books Read'
correlation_data = df[['Incomes', 'How many books did you read during last 12months?', 'Race', 'Education']]
correlation_data.rename(columns={'How many books did you read during last 12months?': 'Books Read'}, inplace=True)

# Calculate correlation matrix
correlation_matrix = correlation_data.corr()

# Plot heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f', cbar=True)
plt.title('Correlation Heatmap: Income, Books Read, Race, Education')
plt.show()
```

Results:

- Books Read & Income
 - Little to no positive correlation
- Books Read & Race
 - Very little positive correlation, basically 0
- Books Read & Education
 - Slight positive correlation
 - Strongest of 3 categories
 - Suggests there may be correlation between education & books read





05

Social/Community Impact

Impact of this Study

- Although our findings suggested that there wasn't a lot of correlation between the various factors and numbers of books read, the outcome would probably be much different in a larger and more diverse group of participants
 - We know this because of trends in real life between things such as education and reading (ie. higher education usually indicates increased reading levels)
- In a bigger scope our study could:
 - Provide valuable insights of how disparities in reading across different groups in society come to appear
 - Be used to find communities that need more support when it comes to literacy and reading
 - Find trends of how your demographics have an effect on your reading habits





Thank you!

